

UNIVERSIDAD DE INVESTIGACIÓN DE TECNOLOGÍA EXPERIMENTAL YACHAY

ESCUELA DE CIENCIAS MATEMÁTICAS Y COMPUTACIONALES

Data Bases

Workshop

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1 Database System for Multitemporal Analysis of Andean Ecosystems Using Sentinel-2 Data

Description

Monitoring ecological dynamics in the Ecuadorian Andes is a challenging task due to the complex interaction between climate variability, mountainous terrain, and highly diverse ecosystems such as páramo and Andean forest. Satellite missions such as Sentinel-2 provide multispectral imagery that allows the extraction of spectral bands and vegetation indices (e.g., NDVI, NDMI, BSI) useful for characterizing vegetation structure and environmental conditions.

However, these data are typically processed in isolated workflows and lack an integrated system capable of storing and organizing multitemporal observations. In particular, when analyzing the same geographic polygon across multiple periods of time, the volume and complexity of spectral information increase significantly. Without a well-designed database structure, managing these datasets becomes inefficient and limits the ability to perform temporal analysis.

This project proposes the design and implementation of a database system that stores multispectral satellite information, vegetation indices, geographic locations, and biome classifications across different time periods. The system will allow researchers to retrieve historical spectral signatures for the same spatial region and construct time series datasets that support statistical and machine learning models.

By organizing multitemporal satellite observations in a structured database, the system will facilitate ecological monitoring and enable predictive modeling of vegetation dynamics. In particular, time series models can be applied to detect ecosystem changes, monitor vegetation health, and forecast potential biome transitions under environmental variability. The database therefore serves as the foundational infrastructure that connects remote sensing data with temporal analytical methods for environmental research.

Requirements

Functional Requirements

- Store multispectral satellite band values and vegetation indices.
- Manage spatial information for polygons and geographic coordinates.

- Record observations for multiple time periods for the same spatial region.
- Allow queries to retrieve spectral data for specific locations and dates.
- Support data extraction for time series analysis and predictive modeling.

Non-Functional Requirements

- Ensure data integrity and consistency across temporal records.
- Provide efficient query performance for large satellite datasets.
- Maintain scalability to accommodate additional time periods and regions.
- Guarantee reliability and reproducibility of stored environmental data.
- Ensure compatibility with Python-based data analysis tools.